

JAK-STAT pathway activation compromises nephrocyte function in a *Drosophila* high-fat diet model of chronic kidney disease

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Yunpo Zhao, Jianli Duan, Joyce van de Leemput, Zhe Han 

Center for Precision Disease Modeling, Department of Medicine, University of Maryland School of Medicine, Baltimore, MD 21201, USA • Division of Endocrinology, Diabetes and Nutrition, Department of Medicine, University of Maryland School of Medicine, Baltimore, MD 21201, USA

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Abstract

Chronic kidney disease is a major healthy issue and is gaining prevalence. Using a *Drosophila* model for chronic kidney disease we show that a high-fat diet (HFD) disrupts the slit diaphragm filtration structure in nephrocytes, the fly functional equivalent of mammalian podocytes. The structural disruption resulted in reduced filtration function in the affected nephrocytes. We demonstrate that a HFD activates the JAK-STAT pathway in nephrocytes, which has previously been linked to diabetic kidney disease. JAK-STAT activation was initiated by increased expression and release of the adipokine, Upd2, from the fat body. This leptin-like hormone is a known ligand of JAK-STAT. Both genetic and pharmacological inhibition of JAK-STAT restored nephrocyte HFD-associated dysfunction. Altogether, our study reveals the importance of the JAK-STAT signaling pathway in the adipose tissue–nephrocyte axis and its contribution to HFD-associated nephropathy. These findings open new avenues for intervention in treating diabetic nephropathy and chronic kidney disease.

Highlights

- High-fat diet (HFD) disrupt nephrocyte slit diaphragm structure and filtration
- HFD releases fat body adipokine, Upd2, which activates JAK-STAT in nephrocytes
- Genetic/pharmacological inhibition of JAK-STAT reverses HFD nephrocyte dysfunction
- JAK-STAT signaling mediates adipose-nephrocyte axis in HFD-associated nephropathy

Impact statement

Using a *Drosophila* model for chronic kidney disease, Zhao et al. show that a high-fat diet induces excretion of a leptin-like JAK-STAT ligand from the fat body. Thus, driving the

adipose-nephrocyte (podocyte equivalent) axis through activated JAK-STAT signaling. These findings link obesity to kidney disease, implicating new avenues for therapeutics.

eLife assessment

This study presents **important** new insights linking obesity to kidney disease using a *Drosophila* model. A series of **compelling** experiments demonstrated that a high-fat diet induces the excretion of a leptin-like JAK-STAT ligand from the fat body, driving the adipose-nephrocyte axis through activated JAK-STAT signaling and subsequently causing a functional defect in nephrocytes. While the combination of genetic tools and pharmacological intervention provides **solid** data and confirms the mechanistic link, the phenotypic analysis is restricted to tracer endocytosis and would benefit from immunofluorescence studies and higher animal numbers.

Introduction

Chronic kidney disease is a prevalent health issue, with an estimated ~12% of people affected worldwide, many of whom are unaware of their condition (Coresh, 2017 [↗](#); US Department of Health and Human Services, Centers for Disease Control and Prevention (Atlanta, GA), 2023 [↗](#)). The gradual loss of kidney function results in excess fluid and buildup of metabolic waste compounds. Clinical symptoms include atherosclerosis, chronic inflammation, malnutrition, and insulin resistance among other metabolic imbalances (Serrano et al., 2023 [↗](#)). Altogether these lead to increased morbidity and mortality associated with chronic kidney failure (GBD Chronic Kidney Disease Collaboration, 2020 [↗](#)). Diabetes, high blood pressure, ageing, obesity, and increased BMI are major risk factors for glomerulopathy and chronic kidney disease (Alizadeh et al., 2019 [↗](#); Berthoux et al., 2013 [↗](#); Bonnet et al., 2001 [↗](#); Coresh, 2017 [↗](#); Hsu et al., 2006 [↗](#); Moorhead et al., 1982 [↗](#); Tsuboi et al., 2013 [↗](#)). Notably, patients with primary kidney disease that are also obese have worsened outcomes (Berthoux et al., 2013 [↗](#)) and following kidney transplantation patients with increased BMI are at greater risk of adverse outcomes (Curran et al., 2014 [↗](#)). with risk increasing risk as BMI increases (Hsu et al., 2006 [↗](#)). The link between dietary fat intake and kidney disease has raised interest into the adipose-renal axis; that is how do bodily fat deposits affect kidney function?

Podocytes from patients with chronic kidney disease contain lipid droplets that store excess fat (Herman-Edelstein et al., 2014 [↗](#); Kimmelstiel and Wilson, 1936 [↗](#)), these cause lipotoxicity by disrupting the mitochondria, as well as endocytosis (Lubojevska et al., 2021 [↗](#)), a process crucial to the kidney filtration structure (Wang et al., 2021 [↗](#)). Studies in rats and mice on a HFD have repeatedly shown that the animals suffer from obesity, diabetes (altered insulin homeostasis), and kidney injury marked by functional (albuminuria; blood accumulation of BUN and creatinine; increased urinary biomarkers of kidney damage) and structural (glomerulopathy with glomerular hypertrophy and focal segmental glomerulosclerosis; fibrosis) deficiencies (Altunkaynak et al., 2008 [↗](#); Ha et al., 2022 [↗](#); Jiang et al., 2005 [↗](#); Kuwahara et al., 2016 [↗](#); Lu et al., 2003 [↗](#); Rangel Silveiras et al., 2019 [↗](#); Ruggiero et al., 2011 [↗](#); Sánchez-Navarro et al., 2021 [↗](#); Sun et al., 2020 [↗](#); Szeto et al., 2016 [↗](#); van der Heijden et al., 2015 [↗](#)). Like in patients with chronic kidney disease, lipid droplets have been observed in the kidneys of HFD rats and mice (Deji et al., 2009 [↗](#); Jiang et al., 2005 [↗](#); Sun et al., 2020 [↗](#); van der Heijden et al., 2015 [↗](#)) and have been associated with dysfunctional cellular systems including oxidative stress, renal inflammation, ER-stress, disruption of mitochondrial dynamics, and impaired autophagy-lysosomal pathway in the cells of the kidneys (Cai et al., 2024 [↗](#); Ha et al., 2022 [↗](#); Kuwahara et al., 2016 [↗](#); C. Li et al., 2016 [↗](#); Lu et al., 2003 [↗](#); Rangel Silveiras et al., 2019 [↗](#); Ruggiero et al., 2011 [↗](#); Sánchez-Navarro et al., 2021 [↗](#); Sun et al.,

2020 [Szeto et al., 2016](#) [van der Heijden et al., 2015](#)). Altogether, these indicate the involvement of both local and systemic changes ([Deji et al., 2009](#)). However, our understanding of the pathways that govern the pathogenic effect of superfluous fat intake on kidney function, as well as how to leverage this knowledge to develop effective therapeutics, remains incomplete.

Recently, an HFD model in *Drosophila* showed lipotoxicity, i.e. lipid droplet accumulation, in the fly nephrocytes ([Lubojeńska et al., 2021](#)). Nephrocytes share many characteristics with human podocytes, including genetics, molecular pathways, and function ([Weavers et al., 2009](#) [Zhang et al., 2013b](#), [2013a](#)). Both cell types contain highly specialized filtration structures known as slit diaphragms that act in concert with the basement membrane; the fly lacuna channel is similar to the urinary space in the mammalian Bowman's capsule; and, many key proteins for podocyte function are likewise essential for nephrocyte function ([van de Leemput et al., 2022](#) [Wang et al., 2021](#) [Weavers et al., 2009](#) [Zhuang et al., 2009](#)). Indeed, fly in vivo nephrocyte models have been successfully used to study a variety of human kidney diseases, including forms of monogenic nephrotic syndrome and steroid-resistant nephrotic syndrome (SRNS) ([Ashraf et al., 2013](#) [Bierzynska et al., 2022](#) [Fu et al., 2017](#) [Gee et al., 2015](#), [2013](#) [Gonçalves et al., 2018](#) [Hermle et al., 2018](#), [2017](#) [Milosavljevic et al., 2022](#) [Odenthal et al., 2023](#) [Paul et al., 2023](#) [Zhao et al., 2019](#) [Zhu et al., 2017](#)). The recent study using the HFD *Drosophila* model recapitulated the ectopic lipid droplets and cellular dysfunction observed in chronic kidney disease and found that the adipose-derived triglyceride lipase protects nephrocyte endocytosis under HFD conditions ([Lubojeńska et al., 2021](#)). Here, we used an HFD *Drosophila melanogaster* model of chronic kidney disease to investigate the role and factors of the adipose-renal axis in chronic kidney disease and identified Upd-activated JAK-STAT signaling as a key component.

Results

High-fat diet (HFD) compromises nephrocyte function

To investigate the effect of superfluous fat consumption on nephrocyte function, newly eclosed flies were fed a normal fat diet (NFD) or a high-fat diet (HFD) for seven days. Then, the flies were subjected to fluorescent dye uptake assays to determine nephrocyte function. Compared to the NFD fed flies, the HFD fed flies showed reduced FITC conjugated albumin (FITC-albumin, 66 kD) fluorescence (**Figure 1A** [and B](#)). Likewise, HFD feeding significantly reduced 10 kD dextran uptake by the nephrocytes (**Figure 1C** [and D](#)). Thus, confirming the HFD model in *Drosophila* for studying kidney disease ([Lubojeńska et al., 2021](#)) and demonstrating that in our hands the consumption of superfluous fat caused nephrocyte uptake dysfunction.

HFD alters nephrocyte morphology

The slit diaphragm (SD) is the fundamental filtration structure of nephrocytes. Thus, given the HFD-induced dysfunctional uptake, we next tested whether the SD is affected by HFD. Therefore we looked at the distribution of polychaetoid (Pyd), the fly homolog of human tight junction protein 1 (TJP1, alias ZO-1) and a key component of the SD filtration unit ([van de Leemput et al., 2022](#)). Immunostaining with anti-Pyd antibody showed predominant membrane localization of Pyd in nephrocytes from NFD fed flies. At the cortical surface of NFD nephrocytes, Pyd showed the fingerprint-like pattern characteristic of the SD (**Figure 2A** [and B](#)). However, in the HFD fed flies, the SD fingerprint-like pattern appeared irregular and Pyd showed an uneven distribution with spots of increased signal intensity (**Figure 2A** [and B](#)). These might correlate to the trails of Pyd seemingly hanging from the cortical surface inside the HFD nephrocytes, along with cytosolic accumulation of Pyd protein (**Figure 2A** [and B](#)). These findings show that a HFD leads to structural disruption of the SD filtration unit in nephrocytes, that likely accounts for the nephrocyte functional deficit observed earlier.

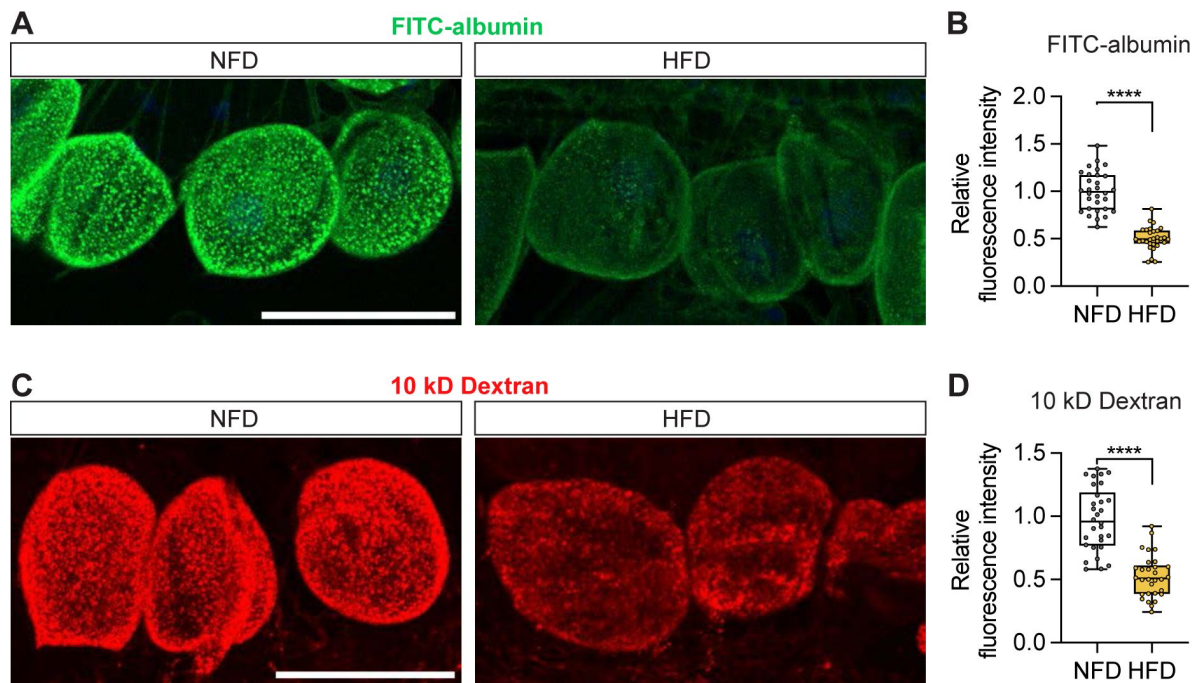


Figure 1.

High-fat diet compromises nephrocyte function

Nephrocytes from control *Drosophila* (w^{1118} , 1-day-old females) fed a regular diet (normal fat diet, NFD) or high-fat diet (NFD supplemented with 14% coconut oil, HFD) for 7 days from eclosion. **(A)** Representative confocal images of nephrocytes show green fluorescence indicative of FITC-albumin uptake. Scale bar: 50 μm . **(B)** Box plot shows the quantitation of the relative fluorescence intensity of FITC-albumin shown in (A); middle line depicts the median and whiskers show Tukey. Statistical analysis was performed with a two-tailed Student's t-test; ****, $P < 0.0001$; $n = 30$ nephrocytes from six 1-day-old females (5 nephrocytes/fly). **(C)** Representative confocal images of *Drosophila* nephrocytes (w^{1118} , 1-day-old females) show red fluorescence indicative of 10 kD dextran uptake. Scale bar: 50 μm . **(D)** Box plot shows the quantitation of the relative fluorescence intensity of 10 kD dextran shown in (C); middle line depicts the median and whiskers show Tukey. Statistical analysis was performed with a two-tailed Student's t-test; ****, $P < 0.0001$; $n = 30$ nephrocytes from six 1-day-old females (5 nephrocytes/fly).

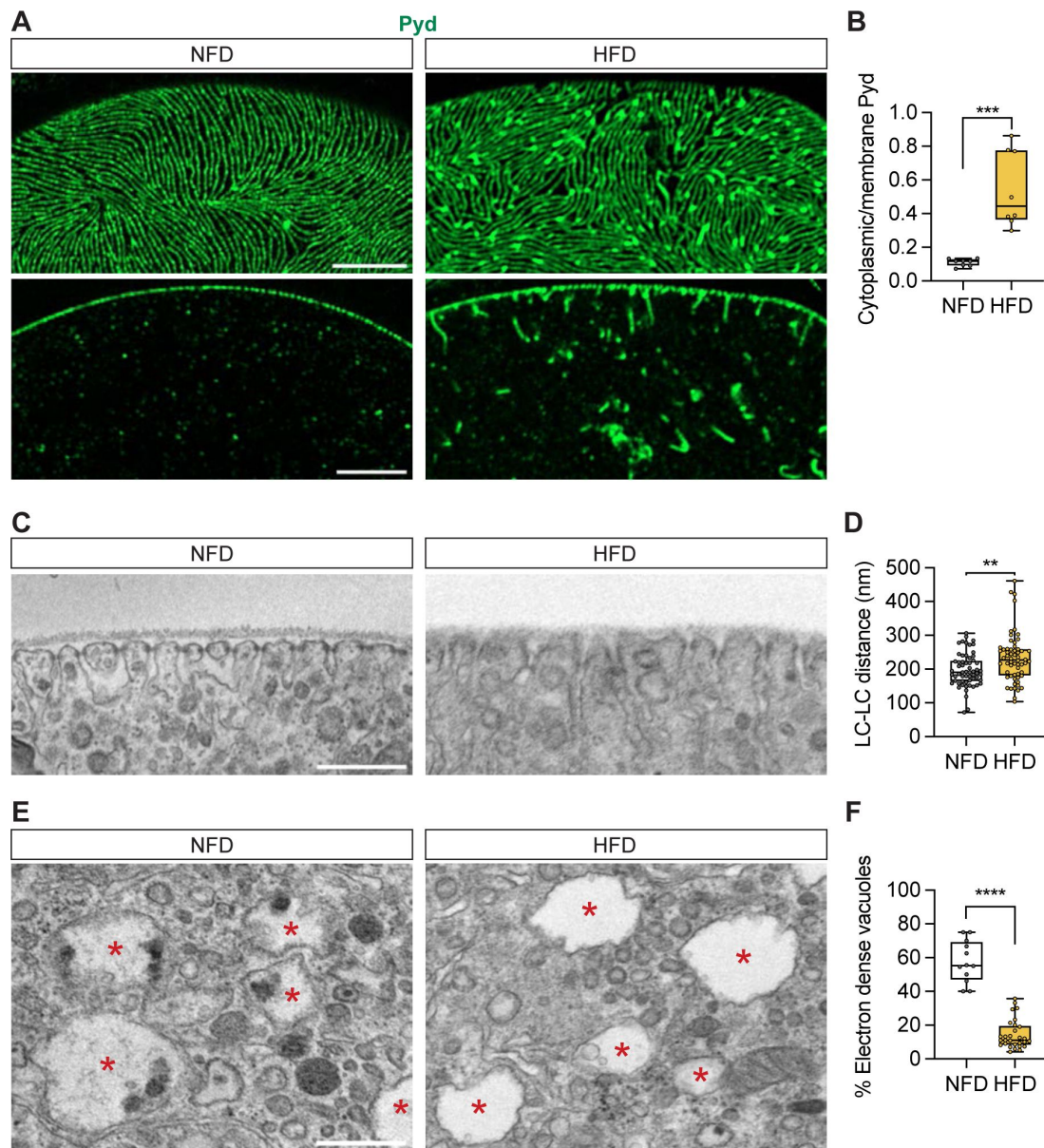


Figure 2.

High-fat diet changes nephrocyte morphology

Nephrocytes from control *Drosophila* (w^{1118} , 7-day-old females) fed a regular diet (normal fat diet, NFD) or high-fat diet (NFD supplemented with 14% coconut oil, HFD). **(A)** Representative confocal images of *Drosophila* nephrocytes immunostained with anti-polychaetoid (Pyd) in green. Upper panels show cortical surface; Scale bar: 5 μ m. Lower panels show subcortical regions; Scale bar: 5 μ m. **(B)** Quantitation of Pyd protein distribution (cytoplasmic vs membrane); middle line depicts the median and whiskers show Tukey. Statistical analysis was performed with a two-tailed Student's t-test; ***, $P < 0.001$; $n = 8$ nephrocytes (1 nephrocyte/fly) from 7-day-old female flies. **(C)** Transmission electron microscopy (TEM) images of *Drosophila* nephrocyte (w^{1118} , 7-day-old females) cortical regions. Scale bar: 0.5 μ m. **(D)** Quantitation of lacuna channel (LC)-LC distance based on images in (C); middle line depicts the median and whiskers show Tukey. Statistical analysis was performed with a two-tailed Student's t-test; **, $P < 0.01$; $n = 60$ LC-LC distance measurements obtained in 10 nephrocytes from six 7-day-old female flies for each group. **(E)** TEM images of *Drosophila* nephrocyte (w^{1118} , 7-day-old-females) cytoplasmic regions. Red asterisks indicate large vacuoles. Scale bar: 0.5 μ m. **(F)** Quantitation of the vacuoles that contain electron dense structures based on images in (E). The middle line depicts the median and whiskers show Tukey. Statistical analysis was performed with a two-tailed Student's t-test; ****, $P < 0.0001$; $n = 12$ nephrocytes for NFD and 29 nephrocytes for HFD from six 7-day-old female flies.

To study the nephrocyte SD and cytoplasmic regions in more detail, we performed transmission electron microscopy (TEM). The nephrocytes of NFD fed flies showed regularly arranged lacuna channels along the cortical surface (**Figure 2C**). The distance between the lacuna channels was significantly increased in nephrocytes from HFD fed flies (**Figure 2C** and **D**). Nephrocytes from both NFD and HFD fed flies showed big subcellular vacuoles (**Figure 2E**). However, the vacuoles in NFD nephrocytes frequently contained electron dense structures, whereas most vacuoles in HFD nephrocytes were clear (**Figure 2E** and **F**). Altogether, these findings demonstrate that a HFD causes significant changes to the nephrocyte and its SD, both at the cortical and subcortical levels.

HFD potentiates the JAK-STAT pathway in nephrocytes

Activity of the Janus kinase/signal transducer and activator of transcription (JAK-STAT) pathway has been linked to diabetic kidney disease (Berthier et al., 2009), likewise it is activated systemically in *Drosophila* in response to a chronic lipid-rich diet (Woodcock et al., 2015). Key components of the JAK-STAT pathway are conserved in fly, including *Signal-transducer and activator of transcription protein at 92E* (*Stat92E*), *hopscotch* (*hop*), *domeless* (*dome*), and *Suppressor of cytokine signaling at 36E* (*Socs36E*) (**Figure 3A** and **B**). Therefore, we tested whether the JAK-STAT pathway was activated in the nephrocytes of HFD-fed flies. We used a fluorescent reporter *10xStat92E-GFP* of JAK-STAT pathway activity (Ekas et al., 2006). Low levels of *Stat92E-GFP* fluorescence were detected in the nephrocytes of flies on a regular diet (NFD). However, in nephrocytes from HFD-fed flies, the fluorescence was significantly increased (**Figure 3C** and **D**). These findings indicate that the consumption of superfluous fat indeed activates the JAK-STAT pathway in nephrocytes.

Activation of the Janus kinase, Hop, reduces nephrocyte function

Since JAK-STAT signaling has been shown to play an important role during development (Brown et al., 2001; Liu et al., 2009), we wanted to isolate its requirement for mature nephrocyte filtration function. To do this we used a dominant gain-of-function allele of *hop* (*Tumorous-lethal, hop.Tum*); a single amino acid change (Gly341Glu) whose expression leads to JAK-STAT pathway activation (Harrison et al., 1995). We expressed *UAS-hop.Tum* specifically in mature nephrocytes using a temperature sensitive *Dot-Gal4* driver (*Dot-Gal4; tub-Gal80ts*, referred to as *Dot-Gal4ts*), known as the TARGET system (McGuire et al., 2004), to turn on expression in adult flies specifically as they are switched to a different environmental temperature (**Figure 4A**). Overexpression of *hop.Tum* significantly reduced nephrocyte absorption ability in the adults, as shown with reduced FITC-albumin and 10 kD dextran in *Dot-Gal4ts>UAS-hop.Tum* nephrocytes compared with control nephrocytes (**Figure 4B-E**).

To validate these results and remove between-fly variability, both biological and technical, we also tested the effect of JAK-STAT pathway activation in tissue mosaic clones using Flp-out (Duan et al., 2020) (**Figure 4F**). For this technique first instar larvae are exposed to heat shock to activate the Hsp70P transcription factor, which induces *Flippase* (*Flp*). Flp activity in turn, removes the stop-cassette thereby freeing *Gal4*. *Gal4* drives the expression of *UAS-GFP* and *UAS-hop.tum* within the same cell. Thus, *Gal4* and therefore *hop.Tum* is only expressed in GFP-labelled nephrocyte clones; whereas, GFP-negative nephrocytes are indicative of no *Gal4* being expressed and serve as internal control cells (**Figure 4F**). In line with the TARGET assays, *UAS-hop.Tum* overexpression significantly reduced nephrocyte absorption ability in adult flies (**Figure 4G** and **H**). These data support a role for JAK-STAT pathway in nephrocyte function.

A

Human Gene	<i>Drosophila</i> Homolog	DIOPT Score	Function
<i>STAT5B</i>	<i>Stat92E</i>	13	transcription factor
<i>PTPRQ</i>	<i>dome</i>	2	transmembrane receptor
<i>JAK1/JAK2</i>	<i>hop</i>	11	non-receptor tyrosine kinase
<i>SOCS4/SOCS5</i>	<i>Socs36E</i>	7	negative regulator of JAK-STAT

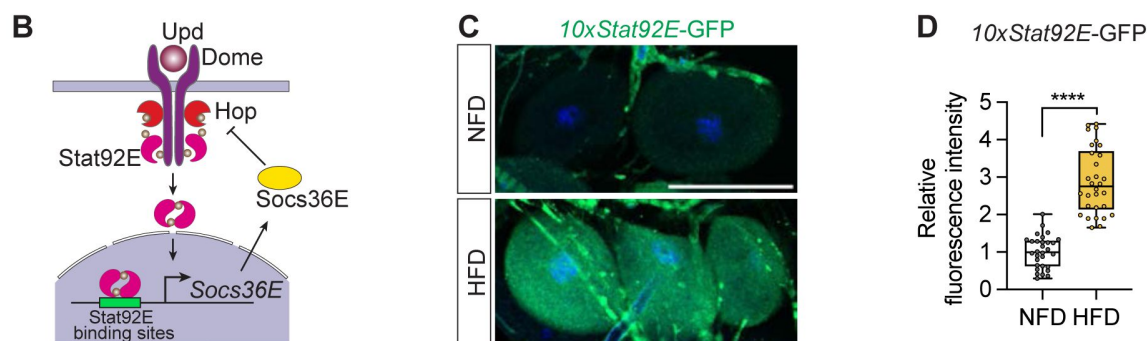


Figure 3.

High-fat diet activates the JAK-STAT pathway in nephrocytes

(A) Table lists human genes encoding JAK-STAT pathway components, along with their *Drosophila* homologs, the DRSC Integrative Ortholog Prediction Tool (DIOPT) score (maximum score = 15), and their function. **(B)** Graphical representation of the JAK-STAT signaling pathway and interaction between its components. Domeless, Dome; JAK Hopscotch, Hop; Signal-transducer and activator of transcription 92E, Stat92E; Suppressor of cytokine signaling at 36E, Socs36E; Unpaired, Upd. **(C)** Representative confocal images of nephrocytes from control *Drosophila* (10xStat92E-GFP, 1-day-old females) fed a regular diet (normal fat diet, NFD) or high-fat diet (NFD supplemented with 14% coconut oil, HFD). 10xStat92E-GFP is shown in green fluorescence; DAPI (blue) stains DNA to visualize the nucleus. Scale bar: 50 μ m. **(D)** Box plot shows the quantitation of the relative fluorescence intensity of 10xStat92E-GFP based on images in (C); middle line depicts the median and whiskers show Tukey. Statistical analysis was performed with a two-tailed Student's t-test; ****, $P < 0.0001$; $n = 30$ nephrocytes from six 1-day-old females (5 nephrocytes/fly).

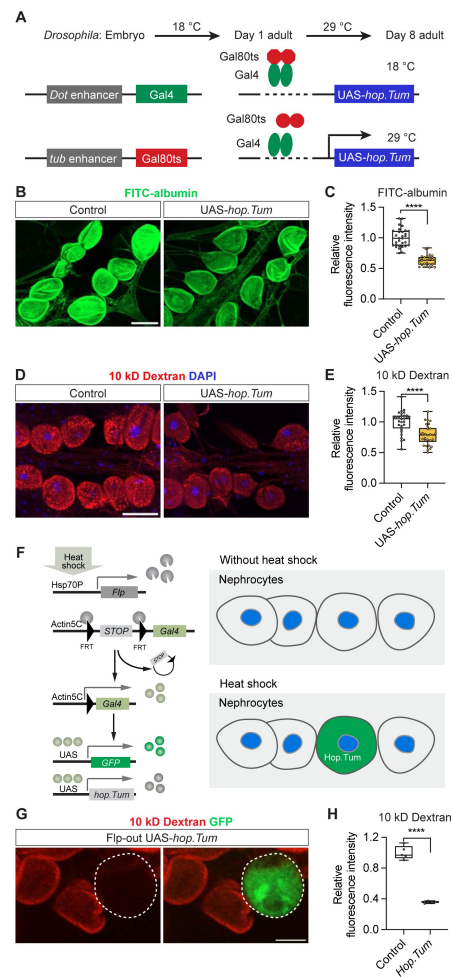


Figure 4.

JAK-STAT pathway activation compromises nephrocyte function

(A) Schematic illustration of targeted UAS-*hop.Tum* expression in the nephrocytes; *hopscotch.Tumorous-lethal*, dominant gain-of-function, constitutively activates JAK-STAT. Temperature sensitive Gal80ts binds to Gal4 and acts as a negative regulator of the Gal4 transcriptional activator at 18°C. A temperature switch to 29°C releases Gal80ts inhibition as it can no longer bind Gal4, thus allowing UAS-*hop.Tum* expression driven by Gal4 to occur. Timeline for temperature switches of the fly at different stages of development have been indicated. **(B)** Representative confocal images of FITC-albumin fluorescence (green) in nephrocytes from control flies (*Dot-Gal4/+; tub-Gal80ts/+*) and those with activated JAK-STAT (*Dot-Gal4/UAS-hop.Tum; tub-Gal80ts/+*). Scale bar: 50 μ m. **(C)** Box plot shows the quantitation of the relative fluorescence intensity of FITC-albumin based on images in (B); middle line depicts the median and whiskers show Tukey. Statistical analysis was performed with a two-tailed Student's t-test; ****, $P < 0.0001$; $n = 30$ from six 1-day-old females (5 nephrocytes/fly). **(D)** Representative confocal images of 10 kD dextran fluorescence (red) in nephrocytes from control flies (*Dot-Gal4/+; tub-Gal80ts/+*) and those with activated JAK-STAT (*Dot-Gal4/UAS-hop.Tum; tub-Gal80ts/+*); DAPI (blue) stains DNA to visualize the nucleus. Scale bar: 50 μ m. **(E)** Box plot shows the quantitation of the relative fluorescence intensity of 10 kD dextran uptake based on images in (D); middle line depicts the median and whiskers show Tukey. Statistical analysis was performed with a two-tailed Student's t-test; ****, $P < 0.0001$; $n = 30$ from six 1-day-old females (5 nephrocytes/fly). **(F)** Schematic illustration of the Flp-out clone strategy to induce UAS-*hop.Tum* expression. Heat shock induces the expression of Flp recombinase, which excises a stop cassette to initiate Gal4 expression. Gal4 binding to the upstream activation sequences (UAS) drives the expression of GFP (as a marker for positive Flp-out clones) and UAS-*hop.Tum*. **(G)** Representative confocal images of 10 kD dextran fluorescence (red) in nephrocytes from flies with a GFP labelled Flp-out UAS-*hop.Tum* clone (*hs-Flp¹²²/+; UAS-Flp^{D1}/UAS-hop.Tum; Act5C>CD2>Gal4^S, UAS-mCD8GFP^{LL6}/+*). **(H)** Box plot shows the quantitation of the relative fluorescence intensity of 10 kD dextran fluorescence uptake based on images in (G); middle line depicts the median and whiskers show Tukey. Control (neighbor of Flp-out clone; UAS-*hop.Tum* (clone). Statistical analysis was performed with a two-tailed Student's t-test; ****, $P < 0.0001$; $n = 5$ from four 3-day-old females.

Depletion of Socs36E, a negative regulator of JAK-STAT, or increased adipokine Upd2, a JAK-STAT ligand, result in decreased nephrocyte function

Next, we looked at additional JAK-STAT pathway components in the context of nephrocyte function. Suppressor of cytokine signaling at 36E (Socs36E) is transcriptionally regulated by Signal transducer and activator of transcription 92E (Stat92E) and functions as a negative regulator of JAK-STAT signaling (Callus and Mathey-Prevot, 2002 [↗](#); Karsten et al., 2002 [↗](#)) (**Figure 3B** [↗](#)) and it is expressed in the nephrocytes (**Figure 3A** [↗](#)). Silencing *Socs36E* specifically in nephrocytes (*Dot-Gal4>Socs36E-RNAi*) significantly reduced nephrocyte uptake function as evident in significantly decreased FITC-albumin (**Figure 5A** [↗](#) and **B** [↗](#)) and 10 kD dextran (**Figure 5C** [↗](#) and **D** [↗](#)) fluorescence compared to the control nephrocytes.

In *Drosophila*, the JAK-STAT pathway ligand unpaired (Upd) family is encoded by *upd1*, *upd2*, and *upd3*. These ligands bind to Domeless (Dome), a single-transmembrane receptor, to activate JAK-STAT signaling (**Figure 3B** [↗](#)). Of these, Upd2 is functionally equivalent to human Leptin and is highly expressed in the fat body, the fly's functional equivalent of vertebrate adipose tissue (Hombria et al., 2005 [↗](#); Rajan et al., 2017 [↗](#); Rajan and Perrimon, 2012 [↗](#)). In *Drosophila*, HFD has been shown to upregulate Upd2 expression and secretion (Rajan et al., 2017 [↗](#); Rajan and Perrimon, 2012 [↗](#)). In our model, over-expression of Upd2, specifically in the fat body (*ppl-Gal4 driver*; *ppl-Gal4>upd2-GFP*), significantly reduced 10 kD dextran uptake in the nephrocytes compared to the control (*ppl-Gal4/+*) (**Figure 5E** [↗](#) and **F** [↗](#)), indicating reduced nephrocyte function. Note, since our *upd2* over-expression line contains *upd2-GFP*, we could not use the FITC-albumin uptake assay, as both signals occupy the same imaging channel.

Like the Janus kinase Hop, the negative regulator Socs36E and the ligand adipokine Upd2 are additional JAK-STAT pathway components that are required for nephrocyte function.

HFD-induced nephrocyte functional defects are mitigated by Stat92E-mediated inhibition of JAK-STAT

To determine that JAK-STAT forms a direct link between HFD and nephrocyte dysfunction, we knocked down *Stat92E* (*Stat92E-IR* (Recasens-Alvarez et al., 2017 [↗](#))) in the nephrocytes to disrupt the JAK-STAT signaling pathway (**Figure 3B** [↗](#)). Deficiency for Stat92E did not affect the uptake function (FITC-albumin or 10 kD dextran) of adult nephrocytes in flies fed a NFD (**Figure 6A-D** [↗](#)). However, under HFD conditions, in which nephrocytes from control flies showed significantly reduced uptake, *Stat92E* deficient nephrocytes showed FITC-albumin and 10 kD dextran fluorescence restored to NFD levels (**Figure 6A-D** [↗](#)). These findings demonstrate that Stat92E, and by extension the JAK-STAT pathway, is required for the nephrocyte dysfunction induced by HFD.

HFD-induced nephrocyte functional defects can be attenuated by methotrexate treatment

Methotrexate suppresses STAT activation; it inhibits the phosphorylation of JAK which is necessary for JAK-STAT pathway activation, and was shown to do so without affecting other phosphorylation-dependent pathways (Thomas et al., 2015 [↗](#)). In our model system, methotrexate treatment (10 μ M; 60 min) reduced levels of Stat92E (10xStat92E-GFP) in the nephrocytes (**Supplementary figure S1** [↗](#)). Stat92E is a key component of the pathway, its reduction is indicative of decreased JAK-STAT signaling. Next, we treated flies on NFD or HFD with 10 μ M methotrexate (60 min incubation; ex vivo) to study the effect of pharmacological JAK-STAT inhibition on nephrocyte function. In nephrocytes from control flies on a regular diet (NFD), the methotrexate treatment had no effect on uptake of FITC-albumin or 10 kD dextran (**Figure 7A-**

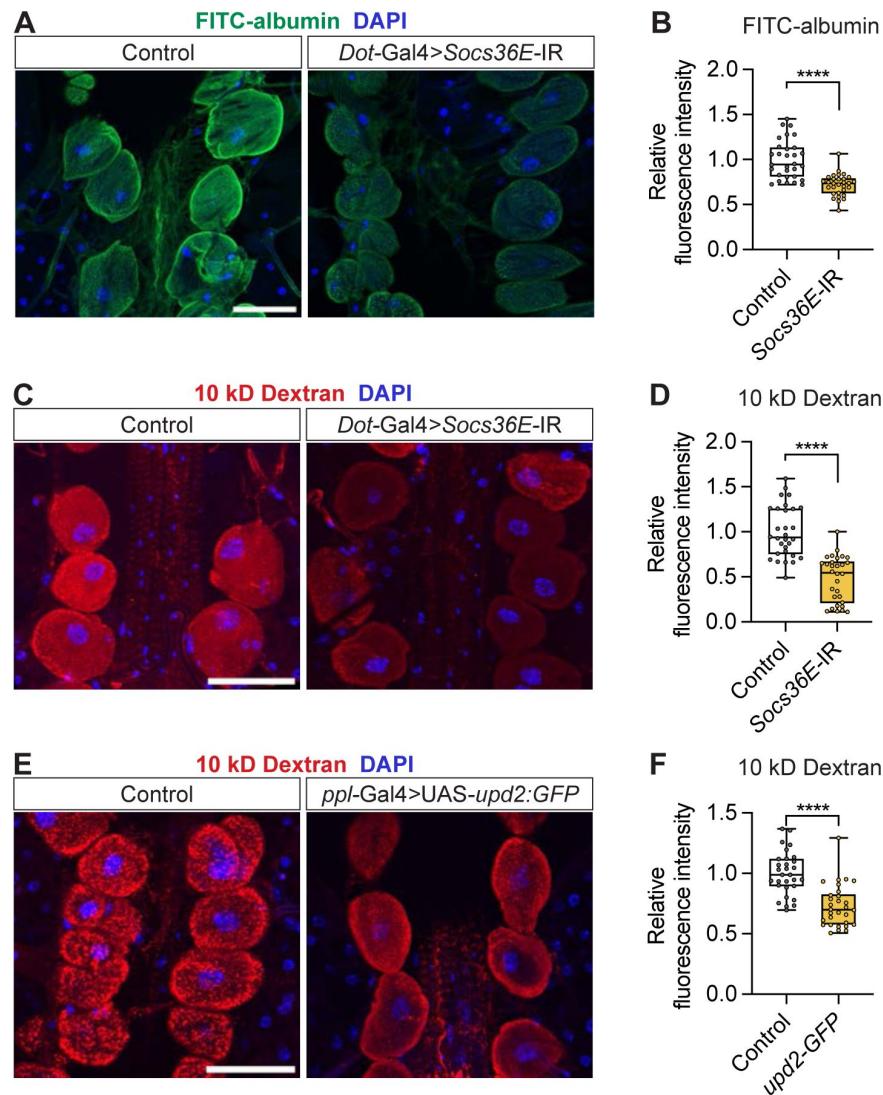


Figure 5.

Silencing *Socs36E* in the nephrocytes, or *upd2* overexpression in the fat body, leads to nephrocyte dysfunction

(A) Representative confocal images of FITC-albumin (green) in nephrocytes from control flies (*Dot-Gal4/+*) and flies with nephrocyte-specific silencing of the *Socs36E* JAK-STAT inhibitor (*Dot-Gal4>Socs36E-IR*); DAPI (blue) stains DNA to visualize the nucleus. Scale bar: 50 μ m. *Socs36E*, *Suppressor of cytokine signaling at 36E*. **(B)** Box plot shows the quantitation of the relative fluorescence intensity of FITC-albumin uptake based on images in (A); middle line depicts the median and whiskers show Tukey. Statistical analysis was performed with a two-tailed Student's t-test; ****, $P < 0.0001$; $n = 30$ from six 1-day-old females (5 nephrocytes/fly). **(C)** Representative confocal images of 10 kD dextran fluorescence (red) in nephrocytes from control flies (*Dot-Gal4/+*) and flies with nephrocyte-specific silencing of the *Socs36E* JAK-STAT inhibitor (*Dot-Gal4>Socs36E-IR*); DAPI (blue) stains DNA to visualize the nucleus. Scale bar: 50 μ m. *Socs36E*, *Suppressor of cytokine signaling at 36E*. **(D)** Box plot shows the quantitation of the relative fluorescence intensity of 10 kD dextran uptake based on images in (C); middle line depicts the median and whiskers show Tukey. Statistical analysis was performed with a two-tailed Student's t-test; ****, $P < 0.0001$; $n = 30$ from six 1-day-old females (5 nephrocytes/fly). **(E)** Representative confocal images of 10 kD dextran fluorescence (red) in nephrocytes from control flies (*ppl-Gal4/+*) and flies with fat body-specific overexpression of JAK-STAT ligand Upd2 (*ppl-Gal4>upd2:GFP*); DAPI (blue) stains DNA to visualize the nucleus. Scale bar: 50 μ m. *ppl*, *pumpless*; *upd2*, *unpaired 2*. **(F)** Box plot shows the quantitation of the relative fluorescence intensity of 10 kD dextran uptake based on images in (E); middle line depicts the median and whiskers show Tukey. Statistical analysis was performed with a two-tailed Student's t-test; ****, $P < 0.0001$; $n = 30$ from six 1-day-old females (5 nephrocytes/fly).

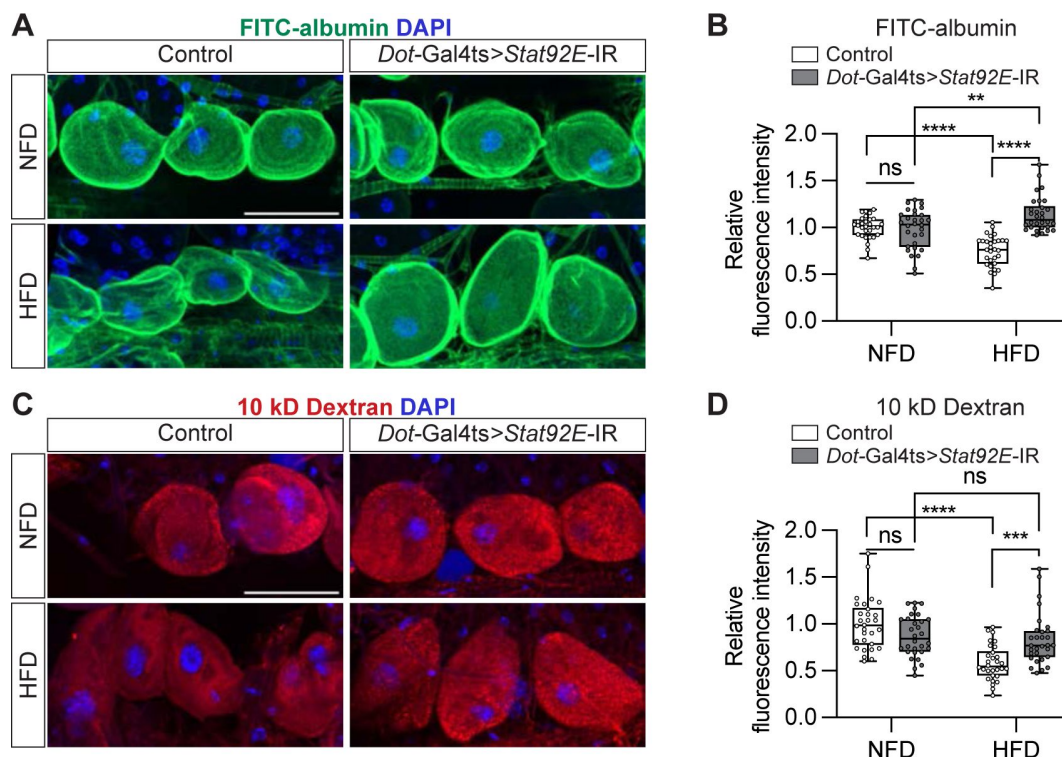


Figure 6.

Silencing *Stat92E* attenuates nephrocyte functional defects caused by a high-fat diet

Nephrocytes from control flies (*Dot-Gal4/+; tub-Gal80ts/+*) and those with *Stat92E* silencing as adults (*Dot-Gal4/UAS-Stat92E-IR; tub-Gal80ts/+*). UAS-*Stat92E*-RNAi expression was induced at the adult stage (see **Figure 4A**) for seven days before the uptake assay. *Stat92E*, *Singal-transducer and activator of transcription 92E*. **(A)** Representative confocal images of FITC-albumin fluorescence (green); DAPI (blue) stains DNA to visualize the nucleus. Scale bar: 50 μ m. **(B)** Box plot shows the quantitation of the relative fluorescence intensity of FITC-albumin uptake based on images in (A); middle line depicts the median and whiskers show Tukey. Statistical analysis was performed by two-way ANOVA with Sidak correction; **, $P < 0.01$; ****, $P < 0.0001$; ns, not significant; $n = 30$ from six 1-day-old females (5 nephrocytes/fly). **(C)** Representative confocal images of 10 kD dextran fluorescence (red); DAPI (blue) stains DNA to visualize the nucleus. Scale bar: 50 μ m. **(D)** Box plot shows the quantitation of the relative fluorescence intensity of 10 kD dextran uptake based on images in (C); middle line depicts the median and whiskers show Tukey. Statistical analysis was performed by two-way ANOVA with Sidak correction; ***, $P < 0.001$; ****, $P < 0.0001$; ns, not significant; $n = 30$ from six 1-day-old females (5 nephrocytes/fly).

D [Figure 7A-D](#)). However, in flies on a HFD, treatment with methotrexate led to significantly increased nephrocyte FITC-albumin and 10 kD dextran uptake, restoring levels to those observed in control fly NFD nephrocytes (**Figure 7A-D** [Figure 6](#) [Figure 6](#)). Thus, like the genetic intervention (*Stat92E* inhibition; **Figure 6** [Figure 6](#)), pharmacological intervention with the JAK-STAT inhibitor methotrexate restored nephrocyte function caused by HFD. These findings support the notion that HFD-induced nephrocyte dysfunction is mediated by the JAK-STAT signaling pathway.

Discussion

A recent publication showed that flies fed a HFD can recapitulate key features of chronic kidney disease, including lipid droplet formation, altered mitochondria dynamics, and endocytosis defects observed as reduced uptake of dextran and albumin (Lubojeńska et al., 2021 [Figure 7E](#)). This previous study found that excess fatty acids, a sign of lipotoxicity, due to a HFD are released from adipose tissue into circulation, then filtered out by the nephrocytes by receptor-mediated endocytosis, at which point the fatty acids accumulate in the lipid droplets (Lubojeńska et al., 2021 [Figure 7E](#)); like those observed in the podocytes of patients with chronic kidney disease (Herman-Edelstein et al., 2014 [Figure 7E](#); Kimmelstiel and Wilson, 1936 [Figure 7E](#)). Notably, lipid droplet lipolysis could counteract the HFD-induced disrupted endocytosis in the nephrocytes via the mitochondria (Lubojeńska et al., 2021 [Figure 7E](#)). This demonstrated one mechanism by which excess lipid droplets, due to HFD, can disrupt nephrocyte kidney function. Here, we likewise used a HFD *Drosophila* model to study chronic kidney disease and revealed another mechanism by which HFD affects both nephrocyte uptake function and morphology of its filtration structure, the slit diaphragm. HFD upregulates the expression of the adipokine Upd2, the functional homolog to human Leptin, which activates the JAK-STAT pathway (Rajan and Perrimon, 2012 [Figure 7E](#)). Our data show this also holds true in the nephrocytes, and as such compromises nephrocyte function (**Figure 7E** [Figure 7E](#)). These findings support obesity as a causal factor in chronic kidney disease.

Of note, in early stage diabetic kidney disease, JAK-STAT pathway genes are upregulated in patient podocytes (Berthier et al., 2009 [Figure 7E](#)). In a rat diabetic model, treatment with a JAK-STAT inhibitor (AG-490) reduced proteinuria (Banes et al., 2004 [Figure 7E](#)); and, overexpression of JAK2 in diabetic mouse podocytes elevated JAK-STAT pathway activity and exacerbated diabetic kidney disease (Zhang et al., 2017 [Figure 7E](#)). These findings, like ours that showed effective treatment of HFD nephropathy using the JAK-STAT inhibitor (methotrexate), support the involvement of the JAK-STAT pathway in HFD-related chronic kidney disease. They also demonstrate that this pathological effect is conserved from flies to mammals. In fact, a phase 2 clinical trial demonstrated that the small molecule baricitinib, a selective JAK1 and JAK2 inhibitor, effectively lowered albuminuria in patients with type 2 diabetes and diabetic kidney disease (Tuttle et al., 2018 [Figure 7E](#)). Altogether these studies support JAK-STAT inhibition as a therapeutic intervention for nephropathy associated with superfluous fat intake (Brosius et al., 2016 [Figure 7E](#)). The conserved disease mechanism makes the HFD fly model a valuable platform to screen JAK-STAT inhibitors for their efficacy to treat chronic kidney disease. The fly findings showed a direct JAK-STAT link at the adipose tissue–nephrocyte axis leads to nephrocyte dysfunction; via HFD upregulated expression of the adipokine Upd2, which activates the JAK-STAT pathway (Rajan and Perrimon, 2012 [Figure 7E](#)) (**Figure 7E** [Figure 7E](#)). In support, like in our fly model, increased leptin has been observed in obese patients (Considine et al., 1996 [Figure 7E](#)) and in patients with chronic renal failure (Heimbürger et al., 1997 [Figure 7E](#)).

Finally, while growing evidence demonstrates that a pharmacological block of the JAK-STAT pathway could effectively treat nephropathy, for decades, the first-line treatment for obesity-related glomerulopathy has been RAS inhibitors (Jiang et al., 2023 [Figure 7E](#)). Their beneficial effects stemming from lowering blood pressure and by lowering the estimated glomerular filtration rate (eGFR) (Banerjee et al., 2022 [Figure 7E](#); Yamout et al., 2014 [Figure 7E](#)). One such inhibitor, telmisartan, targets RAS by blocking the angiotensin (Ang) II receptor and was shown effective in treating nephropathy in a rat model of metabolic syndrome (HFD-fed rats) (H. Li et al., 2016 [Figure 7E](#)). Notably, telmisartan

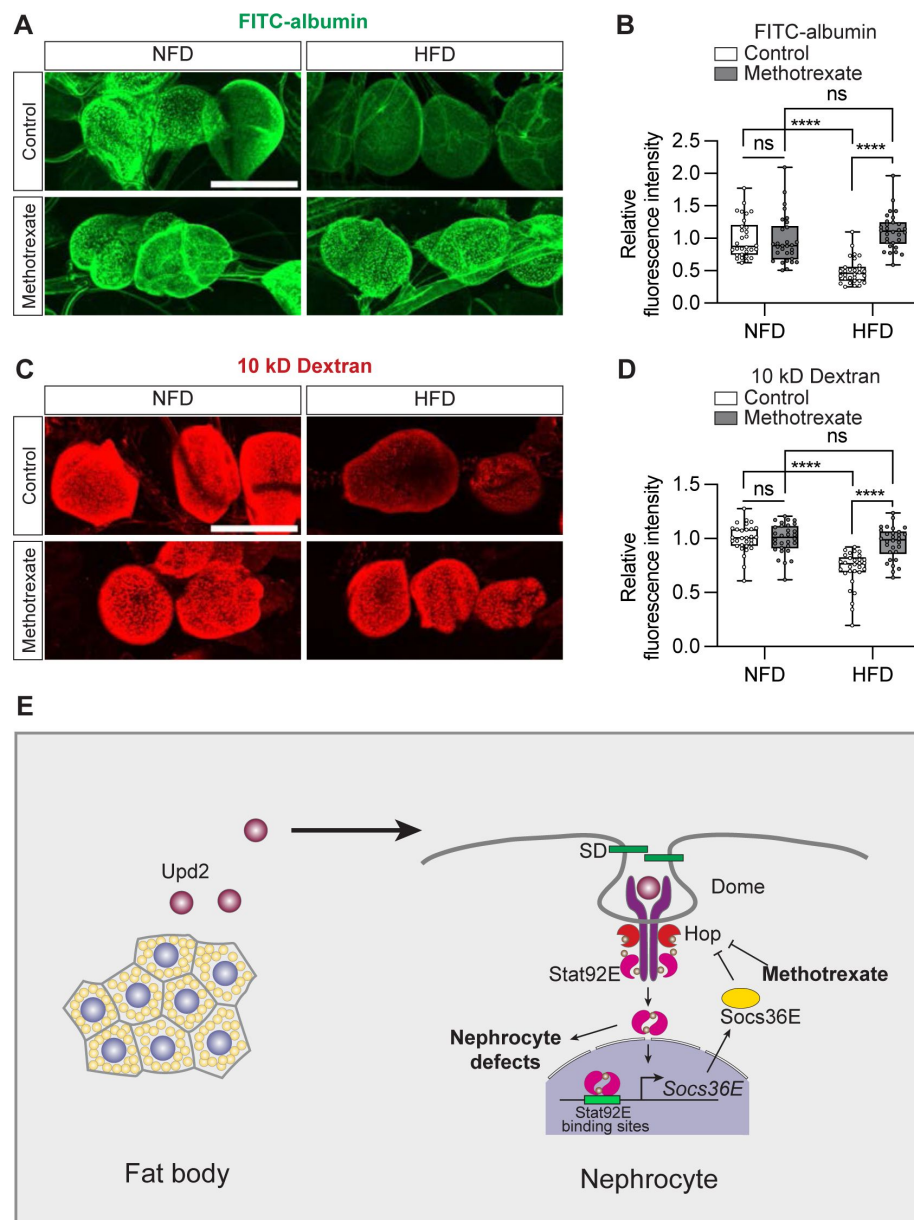


Figure 7.

Methotrexate treatment can restore nephrocyte function following a high-fat diet

Nephrocytes from control *Drosophila* (w^{1118} ; 7-day-old females) fed a regular diet (normal fat diet, NFD) or high-fat diet (NFD supplemented with 14% coconut oil, HFD), with or without methotrexate (10 μ M; ex vivo 60 min) treatment. **(A)** Representative confocal images of FITC-albumin fluorescence (green). Scale bar: 50 μ m. **(B)** Box plot shows the quantitation of the relative fluorescence intensity of FITC-albumin uptake based on images in (A); middle line depicts the median and whiskers show Tukey. Statistical analysis was performed by two-way ANOVA with Sidak correction; ****, $P < 0.0001$; ns, not significant; $n = 30$ from six 1-day-old females (5 nephrocytes/fly). **(C)** Representative confocal images of 10 kD dextran fluorescence (red). Scale bar: 50 μ m. **(D)** Box plot shows the quantitation of the relative fluorescence intensity of 10 kD dextran uptake based on images in (C); middle line depicts the median and whiskers show Tukey. Statistical analysis was performed by Mann-Whitney U test; ****, $P < 0.0001$; ns, not significant; $n = 30$ from six 1-day-old females (5 nephrocytes/fly). **(E)** Graphic of proposed model for high-fat diet-induced nephrocyte defects via an adipose-nephrocyte axis. A high-fat diet upregulates the expression and secretion of the adipokine Unpaired 2 (Upd2), leptin-like hormone, from the fat body. Upd2 is a JAK-STAT ligand, and it activates JAK-STAT signaling at the nephrocytes (Signal-transducer and activator of transcription 92E, Stat92E; Suppressor of cytokine signaling at 36E, Socs36E; JAK Hopscotch, Hop; Domeless, Dome). The overactive JAK-STAT pathway disrupts the integrity of the slit diaphragm (SD) filtration structure and thereby leads to nephrocyte dysfunction.

treatment decreased leptin release from adipose tissue, thereby supporting our model and implicating a hitherto unknown mechanism contributes to leptin-associated nephropathy in metabolic syndrome. In addition, our findings could have implications for lupus nephritis, an inflammatory kidney disease associated with the autoimmune disease systemic lupus erythematosus. Patients with this complication show activated JAK-STAT and elevated leptin, regulated by Sterol regulatory element binding transcription factor 1 (SREBF1) which is involved in lipogenesis (Hao et al., 2013 [↗](#)). Already, phase 2/3 clinical trials are underway to study the efficacy of JAK inhibitors in treating lupus nephritis (Huo et al., 2023 [↗](#)). Our data indicate a possible second pathway could contribute, by direct leptin stimulation of JAK-STAT caused by the dysregulated fat body, warranting further investigation.

Altogether, our study expands our understanding of chronic kidney disease associated with superfluous fat intake and provides new avenues for therapeutic strategies.

Materials and methods

Key resources table

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies		
Chicken anti-GFP	Abcam	Cat. ab13970; RRID:AB_300798
Mouse monoclonal anti-Pyd	Developmental Studies Hybridoma Bank (DSHB)	RRID:AB_2618043
Goat anti-mouse Alexa fluor 488	Invitrogen	Cat. A11029; RRID:AB_2534088
Goat anti-chicken Alexa fluor 488	Invitrogen	Cat. A11039; AB_2534096
Chemicals, Peptides, and Recombinant Proteins		
DAPI	Thermo Fisher Scientific	Cat. D1306
Methotrexate	Sigma-Aldrich	Cas. 06563
10 kD Texas Red-dextran	Thermo Fisher Scientific	Cas. D1828
FITC-albumin solution	Sigma	Cas. A9771
Experimental Models: Organisms/Strains		
<i>Drosophila melanogaster</i> : <i>w</i> ¹¹¹⁸	Bloomington Drosophila Stock Center (BDSC)	RRID: BDSC_3605
<i>Drosophila melanogaster</i> : <i>Dot-Gal4</i>	BDSC	RRID: BDSC_67608
<i>Drosophila melanogaster</i> : <i>ppl-Gal4</i>	BDSC	RRID: BDSC_58768
<i>Drosophila melanogaster</i> : <i>tub-Gal80ts</i>	BDSC	RRID: BDSC_7017
<i>Drosophila melanogaster</i> : <i>10XStat92E-GFP</i>	BDSC	RRID: BDSC_26198
<i>Drosophila melanogaster</i> : <i>Socs36E-IR</i>	BDSC	RRID: BDSC_35036
<i>Drosophila melanogaster</i> : <i>Stat92E-IR</i>	Vienna <i>Drosophila</i> Resource Center (VDRC)	VDRC_106980
<i>Drosophila melanogaster</i> : <i>UAS-hop.Tum</i>	Harrison, et.al., 19955	N/A
<i>Drosophila melanogaster</i> : <i>UAS-upd2:GFP</i>	Hombria, et.al., 2005	N/A
<i>Drosophila melanogaster</i> : <i>hs-Flp</i> ¹²² ; <i>UAS-Flp</i> ^{JD1} /CyO, <i>Act-GFP</i> ^{JMR1} ; <i>Act5C>CD2>Gal4</i> ^S , <i>UAS-mCD8-GFP</i> ^{LL6} /TM6b	Zhao, et.al., 2005	N/A
Software and Algorithms		
FIJI (ImageJ)	Schneider et al., 2012; https://imagej.net/Fiji/Downloads	Fiji-macOS
Adobe Illustrator	https://www.adobe.com/	Adobe Illustrator 2022
GraphPad Prism	https://www.graphpad.com/scientific-software/prism/	GraphPad Prism 9

Drosophila husbandry

Fly lines were reared on a normal fat diet (NFD; Nutri-Fly German formula; Genesee Scientific, San Diego, CA) or a high-fat diet (HFD; NFD supplemented with 14% coconut oil), under standard conditions (25°C, 60% humidity, 12h:12h dark:light cycle), unless otherwise stated.

Drosophila stocks *w*¹¹¹⁸ (BDSC_3605), *Dot-Gal4* (BDSC_67608), *ppl-Gal4* (BDSC_58768), *tub-Gal80ts* (BDSC_7017), *10XStat92E-GFP* (BDSC_26198), and *UAS-Socs36E-RNAi* (BDSC_35036) were obtained from the Bloomington *Drosophila* Stock Center (BDSC). *UAS-Stat92E-RNAi* (VDRC_106980) was obtained from the Vienna *Drosophila* Resource Center (VDRC). *UAS-hop.Tum* was kindly provided by Prof. Norbert Perrimon (Harvard Medical School, Boston, MA; Howard Hughes Medical Institute, Boston, MA). *UAS-upd2:GFP* has been previously described (Hombria et al., 2005) and was kindly provided by Prof. Ylva Engström (Stockholm University, Stockholm, Sweden). The Flp-out line *hs-Flp*¹²²; *UAS-Flp*^{JD1}/CyO, *Act-GFP*^{JMR1}; *Act5C>CD2>Gal4*^S, *UAS-mCD8-GFP*^{LL6}/TM6b was generated previously (Zhao et al., 2015).

FITC-albumin and 10 kD dextran uptake assays

Nephrocyte functional assays were performed ex vivo at room temperature, following a previously described method (Wen et al., 2020) with minor changes. *Drosophila* females were dissected in Schneider's *Drosophila* Medium (Thermo Fisher Scientific, MA), then incubated in a 10 kD Texas Red-dextran solution (0.05 mg/mL; D1828, Thermo Fisher Scientific, MA) in Schneider's *Drosophila* Medium (Thermo Fisher Scientific, MA) for 20 min, or a FITC-albumin solution (10 mM; Sigma, A9771) in Schneider's *Drosophila* Medium (Thermo Fisher Scientific, MA) for 5 min. The specimens were washed in artificial hemolymph twice, followed by fixation in 4% paraformaldehyde (PFA) for 60 min. Then the fixed specimens were washed thrice for 5 min in 1x phosphate buffered saline (1xPBS; pH 7.4) and mounted using Vectashield mounting medium (H-1000, Vector Laboratories, CA). FITC-albumin and 10 kD dextran specimens were imaged using a ZEISS LSM900 confocal microscope with ZEISS Zen acquisition software (blue edition; version 3.0) using a 20× Plan-Apochromat 0.8 N.A. air objective (ZEISS, Oberkochen, Germany). For quantitative comparison of fluorescence intensities, setting for the control condition were chosen to avoid oversaturation (using Range Indicator in ZEN blue; limiting the observed red dots to avoid oversaturation), then applied across the images for all samples/conditions within the assay. The fluorescence intensity of FITC-albumin and 10 kD dextran nephrocytes was determined using Fiji software (Image J (Schneider et al., 2012), version 2.9.0/1.53t; National Institutes of Health, Bethesda, MD). For quantitation, for each genotype, the relative fluorescence intensity of 30 nephrocytes from 6 female flies (5 nephrocytes/fly) was analyzed.

Immunocytochemistry

Immunostaining was performed as previously reported (Zhao et al., 2022) with minor changes. Flies were briefly rinsed in 95% ethanol and dissected in 1xPBS at room temperature. Specimens were incubated in primary antibodies overnight at 4 °C. The incubations in secondary antibodies were performed either overnight at 4 °C or for 2 hours at room temperature. The following antibodies were used: chicken anti-GFP (1:1,000; ab13970, RRID:AB_300798, Abcam, Cambridge, UK), mouse monoclonal anti-Pyd (1:100; RRID:AB_2618043, Developmental Studies Hybridoma Bank, IA), goat anti-mouse Alexa fluor 488 (1:500; A11029, RRID:AB_2534088, Invitrogen, Eugene, OR), and goat anti-chicken Alexa fluor 488 (1:500; A11039, AB_2534096, Invitrogen, Eugene, OR). DAPI (0.5 mg/ml in PBST (0.2% Triton-x 100 in 1xPBS); D1306, Thermo Fisher Scientific, MA) was used to visualize the nuclei. The nephrocytes were imaged using a ZEISS LSM900 confocal microscope (under airyscan mode for Pyd images) with ZEISS ZEN acquisition software (blue edition; version 3.0) and a 63× Plan-Apochromat 1.4 N.A. oil objective (ZEISS, Oberkochen, Germany). For quantitative comparison of fluorescence intensities, setting for the control condition were chosen to avoid oversaturation (using Range Indicator in ZEN blue; limiting the observed red dots to avoid oversaturation), then applied across the images for all samples/conditions within the assay. Image J (Schneider et al., 2012) was used for image processing (version 2.9.0/1.53t; National Institutes of Health, Bethesda, MD). Typically, six flies were imaged per condition, representative images for each are displayed in the figures.

Transmission electron microscopy (TEM)

TEM was performed using standardized procedures. In brief, female adults (7-days-old) were dissected in Schneider's *Drosophila* Medium (Thermo Fisher Scientific, MA). The heart tube and the attached nephrocytes were dissected, removed, and fixed using Sorensen phosphate buffer (2% PFA, 2.5% EM grade glutaraldehyde, 2mM CaCl₂, 0.1 M NaOH; provided by the Electron Microscopy Core Imaging Facility at the Center for Innovative Biomedical Resources (CIBR), University of Maryland School of Medicine, MD). The processed specimens were imaged using a Philips CM100 TEM, carried out at the Electron Microscopy Core Imaging Facility at the Center for Innovative Biomedical Resources (CIBR) (University of Maryland School of Medicine, MD). The LC-LC distances were measured using the Straight Line tool in Fiji (Image J (Schneider et al., 2012 [DOI](#)), version 2.9.0/1.53t; National Institutes of Health, Bethesda, MD) to connect two adjacent lacuna channels (LCs); a straight line was drawn from the middle of the first LC to the middle of the sixth LC, covering five LC-LC intervals on a TEM image, then the measure function was used to obtain the distance value. In total 60 LC-LC distances were measure per condition in 10 nephrocytes from six 7-day-old female flies. The presence of a dark electron dense structure in the vacuoles was manually determined and counted in images obtained from 12 nephrocytes for NFD and 29 nephrocytes from HFD from six 7-day-old female flies.

Tissue mosaic analysis

Flp-out clone (Struhl and Basler, 1993 [DOI](#)) induction was performed as previously described (Duan et al., 2020 [DOI](#)). In brief, female virgins of *hs-Flp*¹²²; *UAS-Flp*^{JD1}/CyO, *Act-GFP*^{JMR1}; *Act5C>CD2>Gal4*^S, *UAS-mCD8-GFP*^{LL6}/TM6b (Zhao et al., 2015 [DOI](#)) were crossed with *UAS-hop.Tum* males. The embryos were collected for 24 hours. First instar larvae (24 hrs after embryo collection) received a 10 min heat shock in a 37°C water bath to induce the mosaic clones. After the heat shock, the larvae were maintained at 25°C. One-day-old female adults were subjected to the 10 kD dextran functional assay (described above). The GFP-positive nephrocyte clones and their neighboring nephrocytes were analyzed.

Methotrexate treatment

Flies were dissected as described for the nephrocyte functional assays. The dorsal cuticle (with nephrocytes) was transferred to methotrexate solution (10 μM; 06563, Sigma-Aldrich, MO) in Schneider's *Drosophila* Medium (Thermo Fisher Scientific, MA) and incubated at room temperature for 60 min. The specimens were rinsed with Schneider's *Drosophila* Medium (Thermo Fisher Scientific, MA), then subjected to a functional assay or fixed in 4% PFA for immunochemistry.

Data analyses and figure preparation

Image J (Schneider et al., 2012 [DOI](#)) (version 2.9.0/1.53t; National Institutes of Health, Bethesda, MD) was used to process the raw data of confocal images and to measure the relative fluorescence intensity. The data sets were tested for normality using the Shapiro-Wilk test and plotted using GraphPad Prism9 software (version 9.5.1). Normally distributed data were analyzed by the two-tailed Student's t-test, or by two-way ANOVA with Sidak correction. Non-normally distributed data were analyzed by Mann-Whitney U test. *P* < 0.05 was considered significant. The figures were arranged using Adobe Illustrator software (version 2022 26.2.1).

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Data availability

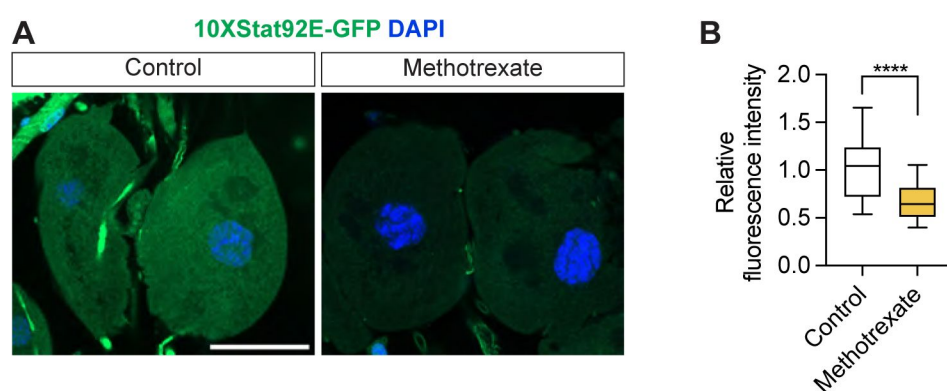
All relevant data can be found within the article and its supplementary information. Requests for resources and reagents related to this manuscript should be directed to and will be fulfilled by the lead contact, Dr. Zhe Han (zhan@som.umaryland.edu).

Author contributions

Z.H. and Y.Z. designed the study; Y.Z. and J.D. carried out the experiments; Y.Z., J.D., J.vdL., and Z.H. analyzed and interpreted the data; Y.Z. prepared the figures; Y.Z., J.vdL., and Z.H. drafted and revised the manuscript; the manuscript has been critically reviewed and the final version approved by all authors. **ORCIDs:** Yunpo Zhao (0000-0002-7942-3406); Jianli Duan (0000-0001-7347-4774); Joyce van de Leemput (0000-0003-1903-7295); Zhe Han (0000-0002-5177-7798)

Disclosures

The authors declare no competing interests.



Supplementary figure S1.

Methotrexate treatment inhibits JAK-STAT pathway activity

(A) Representative confocal images of 1-day-old female adult nephrocytes (*10xStat92E-GFP*). Control, incubated in Schneider's Drosophila Medium (ex vivo for 60 min at room temperature); methotrexate, incubated in 10 μ M methotrexate in Schneider's Drosophila Medium (ex vivo for 60 min at room temperature). *10xStat92E-GFP* in green fluorescence. DAPI staining in blue to visualize the nucleus. Scale bar: 20 μ m. **(B)** Box plot shows the quantitation of the relative fluorescence intensity of *10xStat92E-GFP* based on the images in (A); middle line depicts the median and whiskers show Tukey. Statistical analysis was performed with a two-tailed t-test; ****, $P < 0.0001$; $n = 30$ nephrocytes from six 1-day old female adults (5 nephrocytes/fly).

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Article and author information

Yunpo Zhao

Center for Precision Disease Modeling, Department of Medicine, University of Maryland School of Medicine, Baltimore, MD 21201, USA, Division of Endocrinology, Diabetes and Nutrition, Department of Medicine, University of Maryland School of Medicine, Baltimore, MD 21201, USA

Jianli Duan

Center for Precision Disease Modeling, Department of Medicine, University of Maryland School of Medicine, Baltimore, MD 21201, USA, Division of Endocrinology, Diabetes and Nutrition, Department of Medicine, University of Maryland School of Medicine, Baltimore, MD 21201, USA

Joyce van de Leemput

Center for Precision Disease Modeling, Department of Medicine, University of Maryland School of Medicine, Baltimore, MD 21201, USA, Division of Endocrinology, Diabetes and Nutrition, Department of Medicine, University of Maryland School of Medicine, Baltimore, MD 21201, USA

Zhe Han

Center for Precision Disease Modeling, Department of Medicine, University of Maryland School of Medicine, Baltimore, MD 21201, USA, Division of Endocrinology, Diabetes and Nutrition, Department of Medicine, University of Maryland School of Medicine, Baltimore, MD 21201, USA

For correspondence: zhan@som.umaryland.edu

ORCID iD: [0000-0002-5177-7798](https://orcid.org/0000-0002-5177-7798)

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Reviewer #1 (Public Review):

Summary:

Zhao and colleagues employ *Drosophila* nephrocytes as a model to investigate the effects of a high-fat diet on these podocyte-like cells. Through a highly focused analysis, they initially confirm previous research in their hands demonstrating impaired nephrocyte function and move on to observe the mislocalization of a slit diaphragm-associated protein (pyd). Employing a reporter construct, they identify the activation of the JAK/STAT signaling pathway in nephrocytes. Subsequently, the authors demonstrate the involvement of this pathway in nephrocyte function from multiple angles, using a gain-of-function construct, silencing of an inhibitor, and ectopic overexpression of a ligand. Silencing the effector Stat92E via RNAi or inhibiting JAK/STAT with Methotrexate effectively restored impaired nephrocyte function induced by a high-fat diet, while showing no impact under normal dietary conditions.

Strengths:

The findings establish a link between JAK/STAT activity and the impact of a high-fat diet on nephrocytes. This nicely underscores the importance of organ crosstalk for nephrocytes and supports a potential role for JAK/STAT in diabetic nephropathy, as previously suggested by other models.

Weaknesses:

The analysis is overly reliant on tracer endocytosis and single lines. Immunofluorescence of slit diaphragm proteins would provide a more specific assessment of the phenotypes.

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Reviewer #2 (Public Review):

Summary:

In their manuscript, Zhao et al. describe a link between JAK-STAT pathway activation in nephrocytes on a high-fat diet. Nephrocytes are the homologs to mammalian podocytes and it has been previously shown, that metabolic syndrome and obesity are associated with worse outcomes for chronic kidney disease. A study from 2021 (Lubojemska et al.) could already confirm a severe nephrocyte phenotype upon feeding *Drosophila* a high-fat diet and also linking lipid overflow by expressing adipose triglyceride lipase in the fat body to nephrocyte dysfunction. In this study, the authors identified a second pathway and mechanism, how lipid dysregulation impact on nephrocyte function. In detail, they show activation of JAK-STAT signaling in nephrocytes upon feeding them a high-fat diet, which was induced by Upd2 expression (a leptin-like hormone) in the fat body, and the adipose tissue in *Drosophila*.

Further, they could show genetic and pharmacological interventions can reduce JAK-STAT activation and thereby prevent the nephrocyte phenotype in the high-fat diet model.

Strengths:

The strength of this study is the combination of genetic tools and pharmacological intervention to confirm a mechanistic link between the fat body/adipose tissue and nephrocytes. Inter-organ communication is crucial in the development of several diseases, but the underlying mechanisms are only poorly understood. Using *Drosophila*, it is possible to investigate several players of one pathway, here JAK-STAT. This was done, by investigating the functional role of Hop, Socs36E, and Stat92E in nephrocytes and has also been combined with feeding a high-fat diet, to assess restoration of nephrocyte function by inhibiting JAK-STAT signaling. Adding a translational approach was done by inhibiting JAK-STAT signaling with methotrexate, which also resulted in attenuated nephrocyte dysfunction. Expression of the leptin-like hormone upd2 in the fat body is a good approach to studying inter-organ communication and the impact of other organs/tissue on nephrocyte function and expands their findings from nephrocyte function towards whole animal physiology.

Weaknesses:

Although the general findings of this study are of great interest, there are some weaknesses in the study, which should be addressed. Overall, the number of flies investigated for the majority of the experiments is very low (6 flies) and it is not clear whether the flies used, are from independent experiments to exclude problems with food/diet. For the analysis, the mean values of flies should be calculated, as one fly can be considered a biological replicate, but not all individual cells. By increasing the number of flies investigated, statistical analysis will become more solid. In addition, the morphological assessment is rather preliminary, by only using a Pyd antibody. Duf or Sns should be visualized as well, also the investigation of the different transgenic fly strains studying the importance of JAK-STAT signaling in nephrocytes needs to include a morphological assessment. Moreover, the expected effect of feeding a high-fat diet on nephrocytes needs to be shown (e.g. by lipid droplet formation) and whether upd2 is actually increased here should also be assessed. The time points of assessment vary between 1, 3, and 7 days and should be consistent throughout the study or the authors should describe why they use different time points.

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